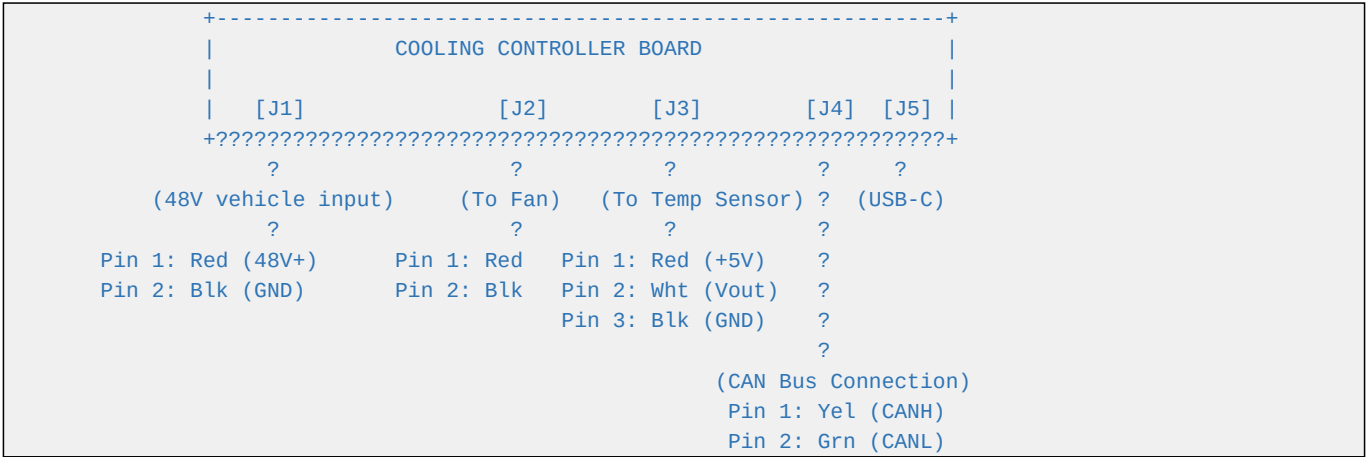


RayGlides 3W/4W EV Kit Cooling Controller - Wiring Diagram

This document details the external wiring harness hookups and connector interfaces.

1. Wiring Harness Layout Diagram



2. Connector Mappings and Pinouts

A. Connector J1: Vehicle Power Input

- * **PCB Header Part**: Molex 43650-0215 (2-pin, single row, SMD, vertical)
- * **Cable Housing Part**: Molex 43645-0200 (2-pin receptacle)
- * **Crimp Terminal**: Molex 43030-0007 (female, tin-plated)
- * **Pin Definitions**:
- * **Pin 1**: +48V DC nominal vehicle traction pack positive line (Red, 20 AWG)
- * **Pin 2**: Ground power return (Black, 20 AWG)

B. Connector J2: Cooling Fan Output

- * **PCB Header Part**: Molex 43650-0215 (2-pin, single row, SMD, vertical)
- * **Cable Housing Part**: Molex 43645-0200 (2-pin receptacle)
- * **Pin Definitions**:
- * **Pin 1**: +12V DC filtered positive rail (Red, 20 AWG)
- * **Pin 2**: Switched GND return (switched by Q1 low-side MOSFET, Black, 20 AWG)

C. Connector J3: Temperature Sensor Extension

- * **PCB Header Part**: Molex 43650-0315 (3-pin, single row, SMD, vertical)
- * **Cable Housing Part**: Molex 43645-0300 (3-pin receptacle)

RayGlides EV Kit Cooling Controller Design File

- * ****Pin Definitions****:

- * ****Pin 1****: +5V DC filtered sensor power rail (Red, 24 AWG)

- * ****Pin 2****: Analog Vout signal return (White or Yellow, 24 AWG)

- * ****Pin 3****: Analog Ground return (Black, 24 AWG)

- * ***Shielding***: Twist the wires and wrap in copper braid shield. Connect the shield braid to ****Pin 3**** at the PCB side only.

D. Connector J4: CAN Bus Output

- * ****PCB Header Part****: Molex 43650-0215 (2-pin, single row, SMD, vertical)

- * ****Cable Housing Part****: Molex 43645-0200 (2-pin receptacle)

- * ****Pin Definitions****:

- * ****Pin 1****: CANH differential data line (Yellow, 24 AWG twisted pair)

- * ****Pin 2****: CANL differential data line (Green, 24 AWG twisted pair)